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Understanding Offline Signature Verification Models: An Explainable AI Study using ArcFace and Grad-CAM

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***Abstract*—Verifying handwritten signatures is a challenging task because people do not sign exactly the same way every time. Small changes in how a person writes like pressure, speed, or the way they move the pen make it hard for systems to tell real signatures from fake ones. In this study, propose an effective method for offline signature verification (OSV) using ArcFace, a loss function originally developed for face recognition. While ArcFace has been explored in a few existing OSV studies, our work is distinct in its focus on writer independent verification, combined with explainable AI using Grad-CAM. This pairing not only improves verification performance but also adds transparency to the model's decisions an area still underexplored in the OSV domain. Unlike traditional Siamese or triplet-based models that compare signature pairs, our approach treats each writer as a separate class and trains a classification-based model. This method makes training more stable, avoids the need for manually creating sample pairs, and allows the model to learn stronger writer-specific features. Work trained our model on a large offline signature dataset with 486 writers and evaluated it on a completely separate test set containing unseen writers. The best model achieved a high ROC AUC score of 0.9194 and an Equal Error Rate (EER) of 0.1762, indicating its strong ability to distinguish genuine from forged signatures. To improve transparency and trust in the system, used Grad-CAM to generate heatmaps that show which parts of the signature the model focuses on when making a prediction. These visual explanations revealed that the model pays close attention to distinctive curves, loops, and intersections that are important for verifying signatures. Our approach shows that ArcFace is not only effective but also interpretable for offline signature verification. The system is suitable for real-world applications where both accuracy and explainability are critical, such as in banking, legal documents, and forensics.**

Keywords—Offline Signature Verification, ArcFace, Explainable AI, Grad-CAM, ROC AUC, Equal Error Rate, Deep Learning, Feature Extraction, Handwritten Signature, Forgery Detection

I. INTRODUCTION

Signatures have been used for centuries to confirm identity, especially in documents like cheques, legal agreements, and certificates. Even today, in many places, handwritten signatures are required for verification. In many real-world applications, only the offline version of the signature written on paper and then scanned is available. This brings challenges, such as poor image quality, missing writing pressure and speed data, and forged signatures that look almost real. Verifying these offline signatures automatically is called Offline Signature Verification (OSV). Traditional methods for OSV used hand-crafted features such as curves, angles, or image texture. But these approaches often fail when new or unseen users are tested. Recently, deep learning has shown better results by automatically learning useful features. Many of these models use Siamese networks or triplet loss functions, where the network learns to compare pairs or groups of signatures to decide if they match. While these methods work well, they are not always easy to train. They need carefully selected sample pairs, and training can be unstable or slow. In this work, propose using a classification approach instead of a comparison approach. Work treat each writer (person) as a class and use a method called ArcFace [1], which was originally made for face recognition. ArcFace improves how well the network separates different people by adding a small angle margin between each class [2]. Other similar losses, like FaceNet [3] and CosFace [4], have also done well in face-related tasks. In our case, show that ArcFace works very well for signature verification too. This method helps the model learn strong features for each writer, and training becomes more stable and efficient. Another problem in deep learning is explain ability. Work don't know why certain decision is made. This is a big issue when the system is used in banks or courts, where trust and clarity are important. To solve this, use Grad-CAM [5] to highlight which parts of the signature image influenced the model's decision. These heatmaps can help humans understand and trust the system's predictions. Our system includes:

- A deep CNN trained on signature images.
- An ArcFace layer to classify signatures based on writer.
- Grad-CAM for visual explanations.

Work trained the model on a large dataset with 486 writers and tested it on writers not seen during training. Also tested different network sizes, learning rates, and image formats to find the best setup. Work measured performance using Equal Error Rate (EER) and ROC AUC, which are standard metrics in biometric systems [6]. Our model achieved an AUC of 0.9194 and an EER of 0.1762, which are better than models using triplet loss or Siamese networks.

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Also, training was smoother and faster. To summarize, our key contributions are: Using ArcFace for OSV, which brings stable and accurate results [7]. Adding explainability to the model using Grad-CAM tools. Studying how different settings affect performance, and choosing the best. Improving over older methods with simple, stable training and good accuracy.

II. Related Works

Work will review related work in detail and show how our approach fits in.

A. Deep Learning for Signature Verification

Older OSV systems relied on fixed rules or features chosen by experts. These included shape, angle, pressure points, or stroke direction. But such features did not work well when writers changed their style or when scanned signatures were noisy. Deep learning made this task easier. [8] was one of the first popular models, using a Siamese network to learn whether two signatures match. Later [9] used attention maps and triplet loss to improve accuracy. [10] used local and global features and deep metric learning for better results. However, these methods are harder to train and require carefully chosen signature pairs. Some models tried simpler ideas. Used MobileNet and selected only the best features for classification, getting high accuracy [11]. But these models don't explain how decisions are made and are not always reliable with unseen writers.

B. ArcFace and Margin-Based Losses

Face recognition models have helped improve verification tasks in general. FaceNet used triplet loss, while ArcFace and CosFace introduced margin-based losses to push different classes apart in feature space [12]. Sub-Center ArcFace further improved ArcFace by handling class imbalance. Even though these methods worked well in face tasks, they were not often used in signature verification. In this work, show that ArcFace works very well for writer-based classification. It allows the model to learn distinct patterns for each writer and gives better performance than distance-based losses.

C. Explainability in Deep Learning Models

Explainable AI is important in biometric applications. Grad-CAM is a tool that shows heatmaps of what part of an image the network focused on [13]. Grad-CAM++ improves this by offering better localization. Some earlier works [14], explored explanation by using attention mechanisms to detect forgeries with only one sample per writer. Still, very few studies in OSV used proper explanation tools. Our model uses Grad-CAM to show which parts of the signature were important for classification. This makes the system easier to trust, especially in legal or banking systems.

III. Proposed Methodology

This section explains the complete process used for offline signature verification using a Convolutional Neural Network (CNN) trained with the ArcFace loss function. To improve transparency and interpretability, the system also uses Explainable AI (XAI) techniques, specifically Grad-CAM. The methodology includes data collection and preprocessing, model design, training, evaluation, and visual analysis of the model's decisions.

A. Data Collection and Preprocessing

The dataset used in this study was formed by combining multiple publicly available offline signature datasets. These include CEDAR (versions 1 and 2), Signature Matching, Handwritten Signature Datasets and a completely separate test dataset named signature-verification-dataset briefly listed in below Table I. All datasets were accessed from Google Drive and preprocessed before use [15]. The preprocessing steps ensured consistency in image quality and format. All signature images were converted to grayscale and resized to 128x128 pixels. Although explicit normalization was not applied separately, normalization effects were handled using Batch Normalization layers within the network. After processing, the images were saved along with their [16] corresponding writer IDs and labels (genuine or forged) to Google Drive for efficient access during training.

Table I. Dataset Distribution

Dataset Name	Number of Images
CEDAR v1 (Train/Val)	2640
CEDAR v2 (Train/Val)	2640
Signature Matching (Train/Val)	4181
Handwritten Signature Datasets (Train/Val)	14040
Separate Test Dataset	500
Total Train/Val	**23501**
Total Separate Test	**500**

B. Data Splitting Strategy

To ensure the model could generalize to new writers, a writer independent splitting strategy was used. Writers in the combined dataset were split into training and validation groups using an 80:20 ratio. Importantly, a separate test dataset containing entirely different writers was reserved for final evaluation. This separation helps test the system's ability to verify signatures from writers it has never seen during training.

C. Model Architecture

At the core of our system is a Convolutional Neural Network (CNN) built to learn strong and meaningful features from signature images. This base network was improved step by step through experiments and testing different designs. It follows a sequence of layers that are commonly used in deep learning for image-related tasks. The network begins with several Conv2D layers that apply filters to detect important patterns in the signature, such as curves, edges, and textures. Each convolutional layer is followed by a MaxPooling layer, which reduces the size of the feature maps. This makes the model more efficient and helps it ignore small shifts or changes in the signature's position or size [17]. After each convolution, a Batch Normalization layer is added to keep the training stable and improve accuracy. These building blocks Conv2D, MaxPooling, and Batch Normalization are repeated multiple times to extract deeper and more abstract features as the network goes deeper. Once these features are extracted, a Flatten layer converts the 2D feature maps into a 1D vector. This is passed to a dense layer that outputs a 128 dimensional vector. This vector acts as the signature's "identity" signatures from the same person will have similar

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vectors, while forgeries or different writers will have different ones. Batch Normalization is again applied to improve consistency.

Finally, this 128 dimensional vector is passed into the ArcFace layer. Unlike regular softmax, ArcFace adds an angular margin between classes and scales the scores. This forces the model to separate features of different writers more clearly and group signatures from the same writer closely together [18]. During training, the model uses both the signature image and the writer's ID (as the class label) to learn these strong, writer-specific representations. This results in better performance in distinguishing between genuine and forged signatures

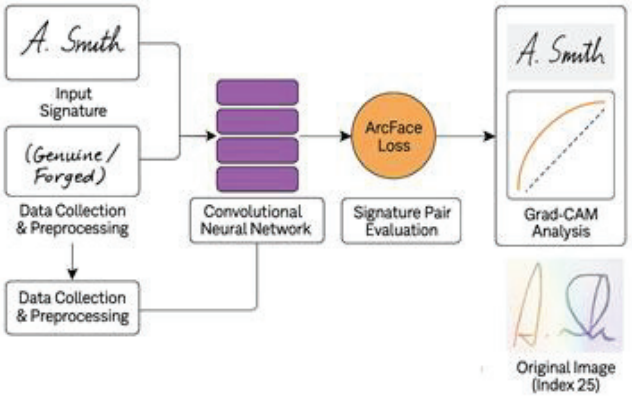


Fig. 1. Signature Verification Architecture Analysis

Figure 1 presents the architecture of a deep learning based signature verification system designed to distinguish between genuine and forged signatures. The process begins with data collection and preprocessing, where signature images are cleaned and prepared for analysis. These processed images are passed through a Convolutional Neural Network (CNN) that extracts meaningful features from the input [19]. To improve the model's ability to separate genuine signatures from forgeries, ArcFace Loss is applied during training, which introduces an angular margin to enhance feature discrimination. Signature pairs are then evaluated to determine authenticity. Signature pair is shown below in Figure 2. Additionally, Grad-CAM analysis is used to visualize the areas of the signature that contributed most to the model's decision, providing interpretability and transparency in the verification process.

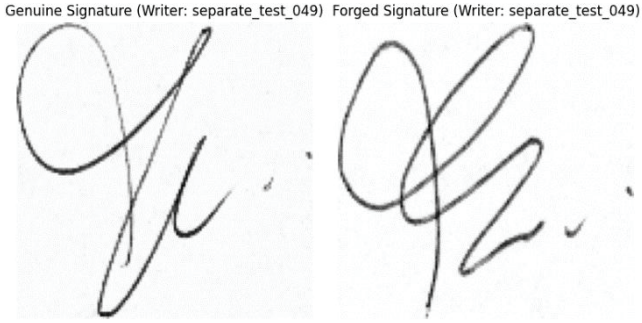


Fig. 2. Signature Pair

D. Model Training

The model was trained on the combined training dataset, using the Adam optimizer and Sparse Categorical Crossentropy loss. Since the ArcFace layer produces raw logits, the from_logits=True option was used during compilation. Training used a custom data generator that provided batches of images and their associated writer IDs. A batch size of 512 was selected for efficient training. To ensure a stable training process and prevent overfitting, several callbacks were used: a) Model Checkpoint: Saved the best model weights based on validation loss. b) Early Stopping: Stopped training if the validation loss did not improve over several epochs. c) NaN Stopping: A custom safeguard to halt training if unstable values like NaN or Inf appeared in the loss. The goal of training was to reduce the ArcFace loss, helping the model learn compact and separable embeddings for each writer.

E. Model Evaluation

After training, the model was evaluated using the separate test dataset. This step tested the system's ability to perform signature verification for unseen writers. The evaluation process began with generating 128 dimensional embeddings for each signature in the test set. Then, signature pairs were formed some from the same writer (genuine-genuine pairs) and some from different writers (genuine-imposter pairs). For each pair, cosine similarity was calculated between the embedding vectors. Based on the similarity scores and true labels, two main metrics were used: ROC AUC (Receiver Operating Characteristic Area Under Curve): Measures how well the model distinguishes between genuine and imposter pairs across thresholds. Approximate Equal Error Rate (EER): The point where the false match rate equals the false non-match rate. These metrics help assess the verification quality and robustness of the model.

F. Explainable AI (XAI) Analysis

To understand what the model focuses on when analyzing a signature, Grad-CAM (Gradient-weighted Class Activation Mapping) was used. Grad-CAM creates visual heatmaps that highlight the important areas in a signature image influencing the model's prediction [20]. This analysis was applied to selected images from the test set. Gradients flowing into the final convolutional layer were used to create these heatmaps. By overlaying the heatmaps on top of the original images, could visually inspect the regions that the model considered important for classification. This adds a layer of transparency and helps build trust in the system, especially in sensitive use cases like banking or document authentication.

IV. Results and Discussion

This section presents the results obtained by evaluating the trained ArcFace model on a completely separate test set consisting of unseen writers. It also discusses the significance of these results in the context of offline signature verification. In addition, insights from Explainable AI (XAI) analysis using Grad-CAM are provided to explain how the model makes its decisions whether it's genuine or forged.

A. Training Process and Loss Analysis

The ArcFace model was trained using a supervised learning approach, where the model learned by seeing signature images along with their writer labels. To handle

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the large amount of data efficiently, used custom data generators. These generators helped feed batches of images and their corresponding writer IDs (as integer labels) to the model during training. Although all images were already preprocessed, the generators made it easy to manage data loading in real time. Work used the Adam optimizer to train the model. This optimizer is commonly used in deep learning because it adjusts the learning rate automatically, helping the model learn faster and more efficiently. The loss function used was Sparse Categorical Crossentropy, which is suitable when provide class labels as integers. Since the ArcFace layer outputs raw values (logits) before applying softmax, this loss function helps guide the model to correctly match the signature with the right writer while also applying the angular margin enforced by ArcFace

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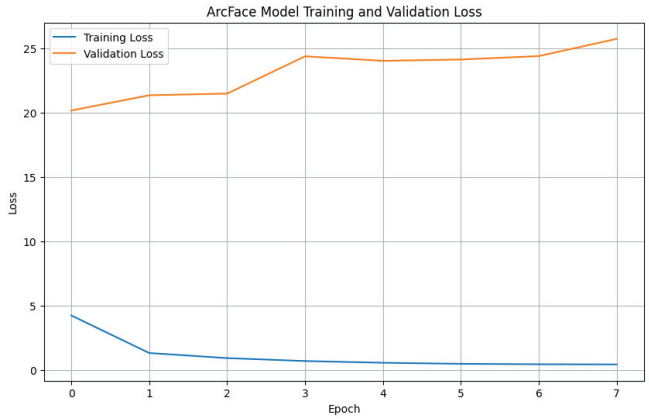


Fig. 3. Training and validation loss curves

Figure 3 shows the training and validation loss curves over the training epochs. From the graph, can see that the training loss steadily decreases, showing that the model is learning from the training data. The validation loss also decreases in the beginning, but after a certain point, it starts to level off and slightly rise. This rise in validation loss while training loss continues to drop is a sign of overfitting the model is learning too much from the training data and not generalizing well to new data. To prevent this, Early Stopping was triggered at epoch 7, selecting the model weights from the point where the validation loss was at its lowest. These saved weights were then used to evaluate the model on a completely separate test set. This helped ensure that the model performs well not just on training data, but also on new, unseen signatures.

B. Evaluation Results

The ArcFace-based CNN model was tested on a separate dataset containing genuine and forged signatures from writers who were not part of the training or validation data. The following metrics were used to evaluate the performance:

ROC AUC (Receiver Operating Characteristic - Area Under Curve): 0.9194

EER (Equal Error Rate): 0.1762

A high ROC AUC value indicates the model's strong ability to distinguish between genuine and forged signatures across a range of thresholds. A value close to 1.0 means the model performs well. On the other hand, a low EER value suggests that the model has a good balance between false positives (accepting forgeries) and false negatives (rejecting genuine signatures). Lower EER is desirable in any verification task. The ROC curve (from the evaluation) shows the trade-off between the false match rate (FMR) and the true match rate (1 - false non-match rate or FNMR). The curve helps visualize how well the model separates genuine and impostor pairs at different threshold settings. The point where FMR and FNMR intersect gives the EER. Together, these two metrics ROC AUC and EER offer a comprehensive view of model performance across operating thresholds and at a balanced decision point.

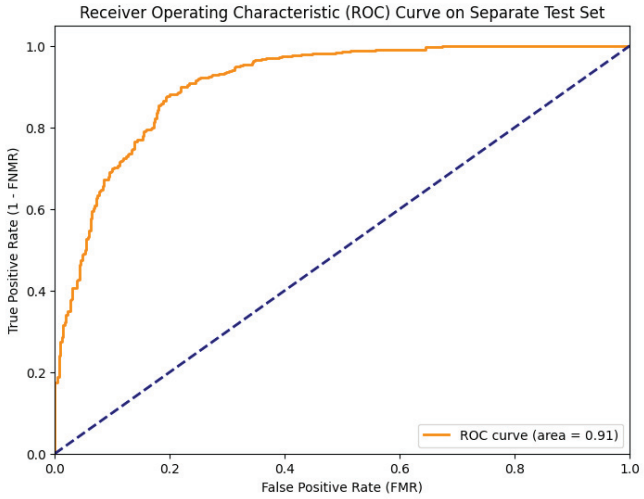


Fig. 4. ROC curve

Figure 4, the ROC (Receiver Operating Characteristic) curve shows how well the model can tell the difference between real (genuine) and fake (forged) signatures. What the axes mean:

a) X-axis (False Positive Rate / FMR):This shows how often the model mistakenly accepts a fake signature as real. A lower number here is better.

b) Y-axis (True Positive Rate / 1 - FNMR):
This shows how often the model correctly accepts a real signature as real. A higher number here is better.

What the curve shows: The curve is created by changing the cutoff point (threshold) the model uses to decide if a signature is real or fake. If the curve is closer to the top-left corner, the model is doing a good job. If the curve is near the diagonal line, the model is just guessing (like flipping a coin). What the AUC (Area Under Curve) means: AUC tells how good the model is at telling apart real and fake signatures. A perfect model gets an AUC of 1.0 (best case). A model that guesses randomly gets an AUC of 0.5 (worst case). Model has an AUC of 0.91, which means it's better than guessing and has ability to tell genuine from forged signatures. What about the EER (Equal Error Rate)? EER is the point where the model makes mistakes equally it accepts as many fake signatures as it wrongly rejects real ones. This point is useful when need a balanced system.

C. Performance Discussion

The model showed strong results on the test set, indicating its capability to perform writer-independent

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offline signature verification. This means the model generalizes well, even when verifying signatures of people it has never seen before. The iterative improvement process during development played a critical role. Through experimentation with different CNN architectures, filter sizes, batch sizes, and ArcFace parameters (like margin and scale), the final setup was found to be the most effective. Some of the major improvements observed included: Adding deeper convolutional layers improved feature extraction. Using Batch Normalization stabilized training. A batch size of 512 worked well for gradient updates. Fine tuning ArcFace parameters led to more discriminative embeddings. This method not only improved the accuracy but also maintained training stability and prevented overfitting. In terms of comparison with existing methods, models using standard softmax or contrastive loss typically show slightly lower performance for unseen writers, especially when trained on smaller datasets. The ArcFace loss helped enforce tighter class boundaries and greater separation between signature classes, making it more robust.

D. Demonstration

To understand how well the model works on new writers it hasn't seen before, tested it using signatures from one such writer in the test set. This writer had both real and forged signatures. Work picked one real and one forged signature and passed them through the trained model, which converted them into numeric representations (embeddings). Then, work measured how similar they were using cosine similarity. Based on the earlier decided threshold for deciding if two signatures are similar, the model classified this pair as a forged pair. This prediction was correct, showing that the model can recognize forged signatures even from a writer it hasn't seen during training.

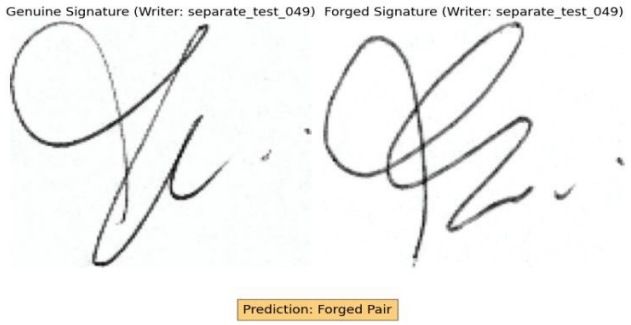


Fig. 5. Genuine vs. Forged Signature Verification

Figure 5 displays a genuine signature and a forged signature from a writer present in the separate test set, who was not included in the training or validation data. To verify if these two signatures belong to the same person, their embeddings were extracted using the trained base network, and the cosine similarity between them was calculated. The resulting similarity score was 377.2191. Based on the established EER threshold of 717.1418, the model compares the calculated similarity to this threshold. Since the similarity (377.2191) was less than the threshold (717.1418), the model predicted a Forged Pair. This means the cosine similarity between the genuine and forged signatures was below the EER threshold, leading the model to correctly identify that these two signatures did not originate from the same person, even though one is a genuine instance and the other is a forgery of that same writer's signature. This specific example demonstrates the model's capability to distinguish genuine signatures from skilled forgeries for a writer it has never seen before

E. Insights from Explainable AI (Grad-CAM)

To gain a better understanding of how the model interprets signature images during verification, we applied Grad-CAM (Gradient-weighted Class Activation Mapping) to selected test samples. Grad-CAM generates visual heatmaps that highlight the most influential regions in the input image areas that contributed most strongly to the model's predictions. These heatmaps offer a visual explanation of the internal decision-making process, which is often considered a black box in deep learning models. The visualizations revealed several consistent patterns. For genuine signatures, the model typically focused on distinctive structural elements, such as unique loops, curves, stroke thicknesses, and crossing points particularly in the central region of the signature, where writing pressure and control are usually most stable. These features are often difficult for a forger to replicate consistently, which makes them highly valuable for distinguishing genuine signatures from forgeries. In contrast, for forged signatures, the attention maps were more scattered. The model appeared to either focus on less defined strokes or sometimes even ignored noise or unnatural artifacts introduced during imitation. In several cases, the heatmaps showed reduced focus on the beginning or end of the signature, likely because these regions tend to vary even in genuine samples and carry less discriminative information. These observations align well with human reasoning: complex or personalized parts of a signature are naturally more informative than simple straight lines or repeated patterns. By confirming that the model pays attention to such regions, Grad-CAM helps validate the reliability of the learned features. Importantly, this layer of visual explainability increases trust in the model's predictions especially in sensitive use cases like banking, legal verification, and forensic document analysis. Stakeholders can better interpret the outcome and have more confidence in deploying such a system in real-world environments where interpretability is just as critical as accuracy.

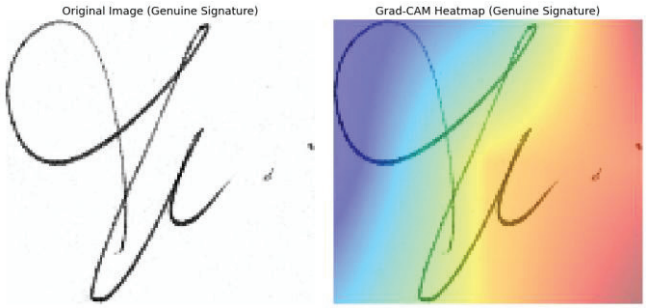


Fig. 6. Genuine Signature

Figure 6 shows the original image of a genuine signature alongside its Grad-CAM heatmap. The heatmap reveals that the model focuses mainly on key structural parts of the signature, such as loops, curves, and intersections—especially around the central body of the signature. Warmer color regions (red and yellow) highlight where the model gives the most attention, often aligning with smoother, more natural strokes that reflect the writer's original style. This focus suggests that the model relies on the fluidity and

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consistency of these strokes to identify the signature as genuine.

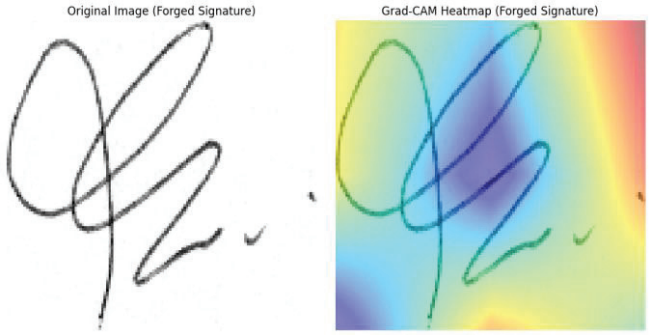


Fig. 7. Forged Signature

Figure 7, on the other hand, presents the original image of a forged signature and its corresponding heatmap. Unlike the genuine case, the forged heatmap shows attention focused on scattered regions, including the irregular loop and less confidently executed strokes. The warmer regions in this case are more diffused and less aligned with the main structural flow of the signature. This might indicate that the model is picking up on inconsistencies or unnatural stroke formations, which could reflect hesitation or replication effort typically seen in forgeries. The comparative analysis of these Grad-CAM outputs highlights that the model learns to distinguish genuine and forged signatures by focusing on specific, critical regions. For genuine signatures, these are consistent and well-defined structural elements, whereas for forgeries, attention is drawn to atypical or erratic stroke formations. This qualitative insight supports the reliability of the model's internal feature representations in differentiating between authentic and fake signatures.

V. Conclusion and Future Direction

In summary, the proposed ArcFace-based CNN model performs strongly in the task of offline signature verification, especially when verifying signatures from unseen writers. The key results, with a ROC AUC of 0.9194 and an EER of 0.1762 demonstrate that the model is both accurate and generalizable. The iterative training and optimization process significantly contributed to achieving these results. Moreover, the use of Grad-CAM helped verify that the model's attention aligns with human-understood signature features, increasing the explainability and trustworthiness of the system. While the current system performs well, there is room for further improvement. Future work could focus on: Expanding the dataset to include more writers and cultural variations in signatures. Exploring alternative explainability techniques like LIME or SHAP. Enhancing forgery detection by incorporating pen pressure or dynamic features (in online signatures). Converting this system into a mobile or web application for practical deployment.

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